

Asia Semiconductors & Global Memory

UMC, Vanguard: Great expectations but on false assumptions?



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UMC and Vanguard have rallied by c. 150% & 65%, respectively, YTD on the thesis of price increase & spillover demand from TSMC (Exhibit 1). Expectations are great, but we find on false assumptions.

“We are increasing our mature node capacity”, said TSMC. We believe TSMC is indeed phasing out its two oldest fabs (Fab 2 (6”) & Fab 5 (8”)), but even if the demand all moves to UMC & Vanguard, it’d represent only 8-13% of their combined 6” & 8” capacity, or 4-6% of their total capacity (Exhibit 2). Moreover, in the most recent earnings call, TSMC confirmed it is expanding mature-node capacity in Japan & Europe. The demand spilled over from TSMC, if any, is hardly meaningful.

Raising prices in 2027 amid high memory cost? UMC & Vanguard successfully raised prices this year & some expect more to come, citing hikes from other foundries. Chinese foundries benefit from localization & PSMC (6770 TT, not covered) sold a fab to Micron. With their fabs fully utilized for a while, their price hikes are well-supported, but these factors don’t apply to UMC & Vanguard. More importantly, “the mature node demand ... is not that strong”, said TSMC and we further worry the demand will be eventually hurt by high memory costs.

Between UMC & Vanguard, we find Vanguard relatively better. TSMC indicated mature-node demand is sluggish, but power management IC & CMOS image sensor are exceptions. With over 75% of its revenue from power management (Exhibit 11), Vanguard is thus relatively better positioned than UMC. Strong CoWoS demand results in strong interposer demand for TSMC, & TSMC outsourcing some of the interposers to Vanguard will help Vanguard bring its new VSMC 12” fab sooner to profitability.

Silicon photonics may be good for UMC but too early to price it in. Though [UMC announced to produce PIC \(photonic IC\) for SILITH not long ago](#), with tech licensing from imec happening just late last year, we think it can take years for UMC to gain traction meaningfully enough to move the needle for the company. Tower Semiconductor (TSEM US, not covered) saw its silicon photonic revenue increase 3x YoY in 1Q26 and has secured contracts for US\$1.3B revenue in 2027. To support that, Tower is expanding capacity with US\$920M capex but we’ve not seen UMC’s capex for this yet.

UMC & Intel may try to collaborate on 3nm, but the chance of success is slim. 3nm will allow UMC to differentiate from Chinese rivals technologically, but without any experience & customer base on 7nm & 5nm, we don’t see what UMC can bring to the collaboration. The capex burden is heavy if UMC wants to share the burden from Intel. And as Intel’s 3nm has been nearly entirely used internally, it will take great efforts to make it “foundry-friendly”. Hence discussion between them is likely but unlikely to materialize into a joint project & financial benefit to UMC.

We updated our models on UMC, Vanguard & Novatek and reiterate Underperform on UMC and Market-Perform on Vanguard & Novatek.

BERNSTEIN TICKER TABLE

Ticker	Rating	27 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				P/B (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
2330.TT (TSMC)	O	TWD	2,350.00	2,780.00	78.1%	TWD	66.25	101.62	124.63	11.2	8.3	6.4
TSM (TSMC)	O	USD	403.41	430.00	47.8%	USD	10.48	16.07	19.70	12.2	9.0	6.9
2303.TT (UMC)	U	TWD	126.00	88.00	162.5%	TWD	3.34	4.58	5.56	4.2	3.8	3.5
OLD				47.00				3.31	3.85			
UMC (UMC)	U	USD	19.48	13.60	150.4%	USD	0.54	0.74	0.89	4.2	3.6	3.4
OLD				7.40				0.53	0.62			
5347.TT (Vanguard)	M	TWD	157.00	146.00	37.2%	TWD	4.30	5.70	6.59	4.6	4.3	4.1
OLD				94.00				4.34	4.44			
3034.TT (Novatek)	M	TWD	518.00	480.00	(22.3)%	TWD	26.87	30.43	32.54	19.3	17.0	15.9
OLD				370.00				24.21	25.10			
ASIA			1,880.30									
SPX			7,411.98									

PRICE TARGET CHANGE / ESTIMATE CHANGE IN BOLD

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

3034.TT valuation is Reported P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

TSMC: We rate TSMC Outperform with PT=NT\$ 2,780.00

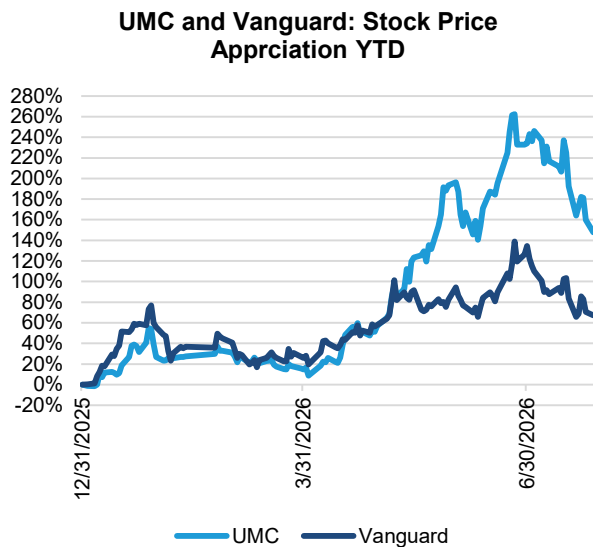
Novatek: We rate Novatek Market-Perform with PT raised to NT\$ 480.00 from NT\$370.00 based on the same 14x target P/E and revised up Q5-Q8 EPS due to better growth in SoC segment.

UMC: We rate UMC Underperform with PT revised up to NT\$ 88.00 from NT\$47.00. We revised up our target P/B to 2.5x from 1.5x as we forecast its ROE to stay in the range of 15-20% going forward, higher vs. the pre-cycle average of below MSD%. We also revised up Q5-Q8 BVPS mainly on higher ASP assumption.

Vanguard: We rate Vanguard Market-Perform with PT raised to NT\$ 146.00 from NT\$94.00 based on revised up 4x target P/B from 3x and revised up Q5-Q8 BVPS mainly on higher visibility for VSMC and better utilization.

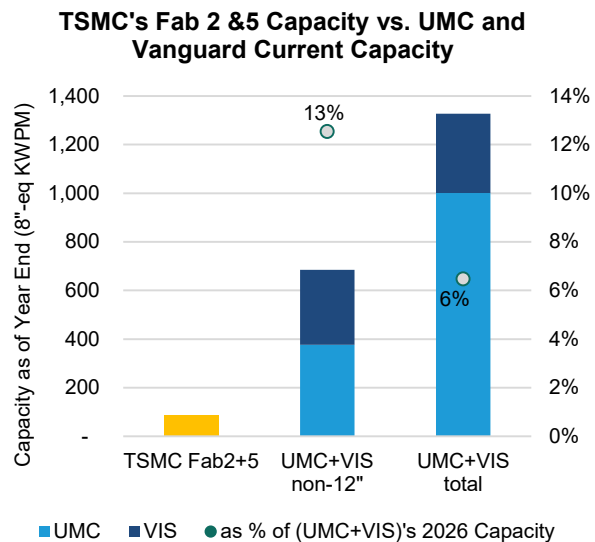
DETAILS

EXHIBIT 1: UMC & Vanguard rallied 148% & 67%, respectively, YTD on the thesis of price increase & spillover demand from TSMC.



Source: Bernstein analysis

EXHIBIT 2: TSMC's Fab 2&5 would represent only 8–13% of combined non-12" capacity and 4–6% of total capacity of UMC and Vanguard.

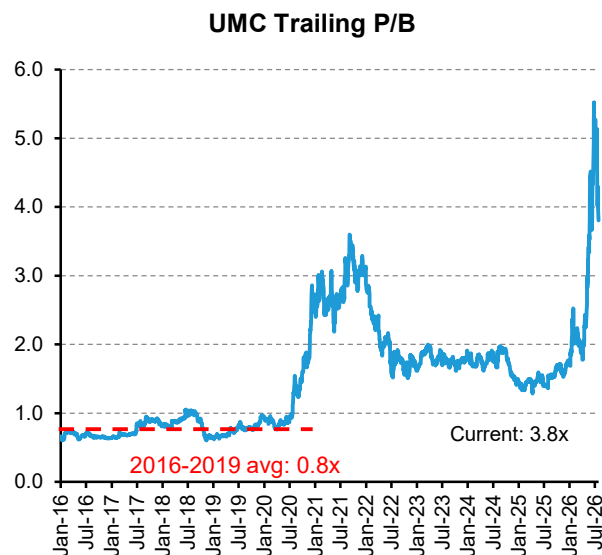


Source: SEMI, Bernstein estimates and analysis

Even after the recent correction, UMC is traded at a peak valuation. We reiterate Underperform.

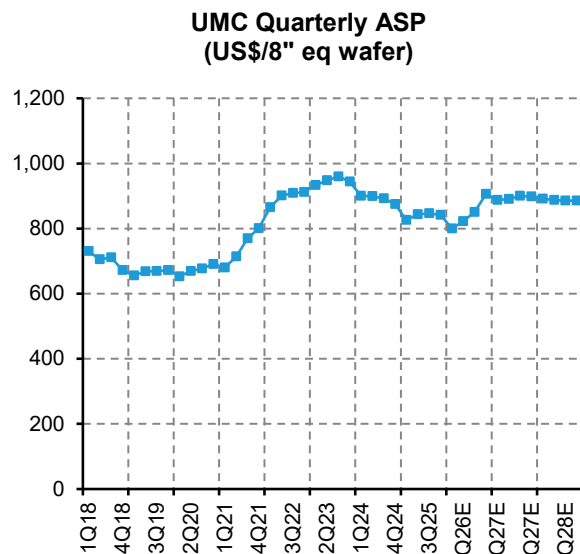
- The stock rallied by c. 150% YTD mainly on the expectation of mature node price hike and spillover demand from TSMC. Speculation on extended collaboration with Intel and CPO business also make some investors interested in the stock. Our channel checks however suggest the price hike thus far is only c. HSD% (High Single-Digit %) in 2H26, but that however is insufficient to justify such a rally. We also don't think the Intel collaboration is likely to take place as UMC does not have any technology or customer base to offer. Silicon photonics may be a market addressable for UMC's mature-node capacity, but it will take years to materialize even if UMC succeeds. The stock is still traded at 3.8x forward P/B even after the recent pullback, still higher than the peak valuation in 2022 when COVID resulted in an unprecedented supply shortage & hence unprecedented price hikes for UMC (Exhibit 3).
- We revised up our ASP assumption for 2026 (Exhibit 4) to reflect the recent price hike & now forecast 3.4% and 6.5% QoQ price increase in 3Q26 and 4Q26, respectively. We also lifted the utilization rate (Exhibit 5) slightly to reflect marginally better supply/demand dynamics for mature nodes. As a result, we forecast UMC's revenue to grow by 16% in 2026 and 15% in 2027, and GM & OPM also on a rising trajectory (Exhibit 6) thanks to better mix (i.e. higher contribution from 22nm). This supports an ROE expansion from 14% to 18% from 2026 to 2028, but that is still a lot lower vs. the ROE of 28% during COVID (Exhibit 7 - Exhibit 8).
- Our projection is below consensus likely because we assume heavier depreciation across 2026-2028 (Exhibit 10). We also assume the price of 8" mature nodes will eventually drop due to competitive pressure from China in the long term. We raise our target P/B to 2.5x from 1.5x on better ROE in the medium term & historical correlation between P/B & ROE (Exhibit 9), and lift our TP to **NT\$88.00** from NT\$47.00. This implies 31% potential downside as of July 24 market close. Dividend yield could provide c. 2% return. Maintain **Underperform**.

EXHIBIT 3: UMC is still trading above the P/B peak in COVID even after recent correction. We believe it's overvalued.



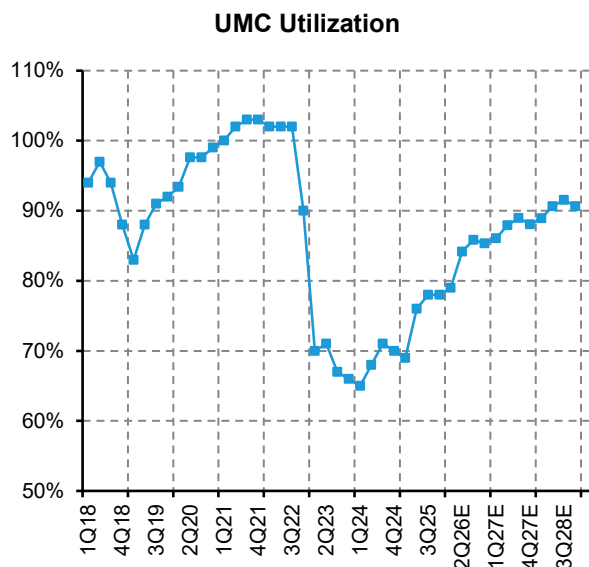
Source: Bloomberg and Bernstein analysis

EXHIBIT 4: We revised up ASP estimates for 2026 to reflect the successful price hike in 2H26. On a longer run, we believe mature node ASP is still under competitive pressure from China.



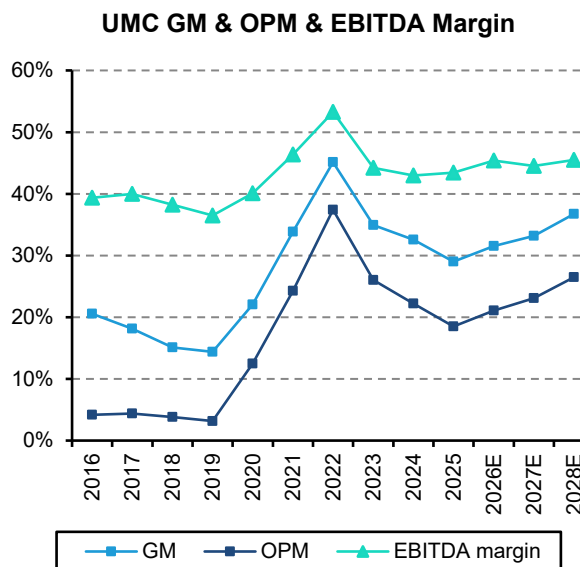
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 5: We also lifted the utilization rate to reflect marginally better supply/demand dynamics for mature nodes.



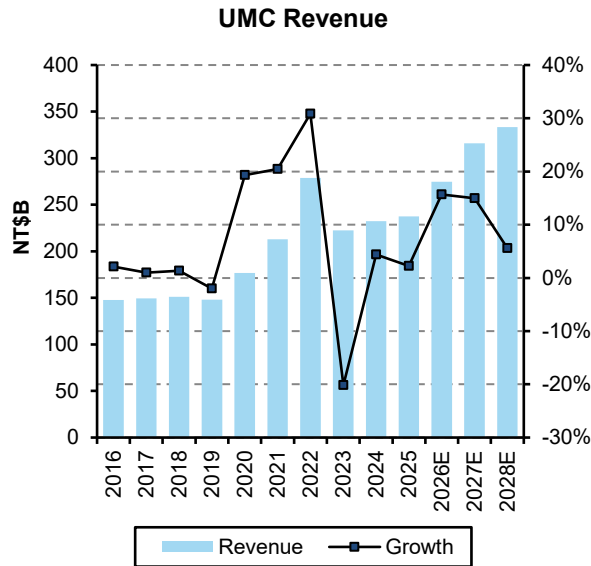
Source: Company report, Bernstein estimates and analysis

EXHIBIT 6: GM & OPM will be on a growth trajectory thanks to a better mix (i.e. higher contribution from 22nm) & higher utilization rate.



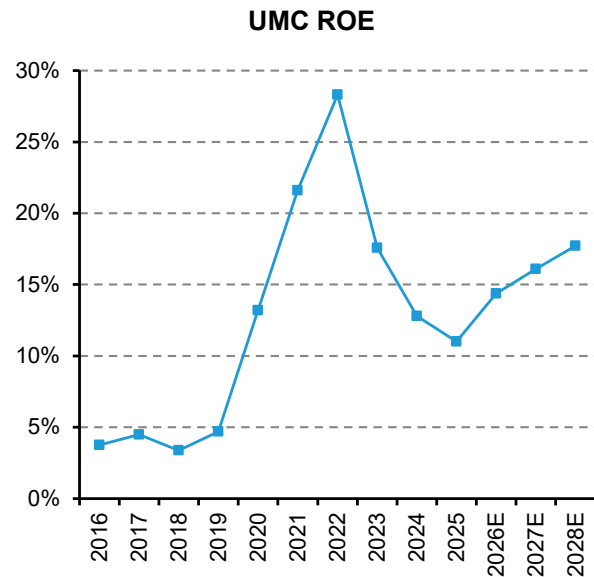
Source: Company report, Bernstein estimates and analysis

EXHIBIT 7: We forecast UMC's revenue to grow by 16% in 2026 and 15% in 2027 driven by higher ASP and utilization rate.



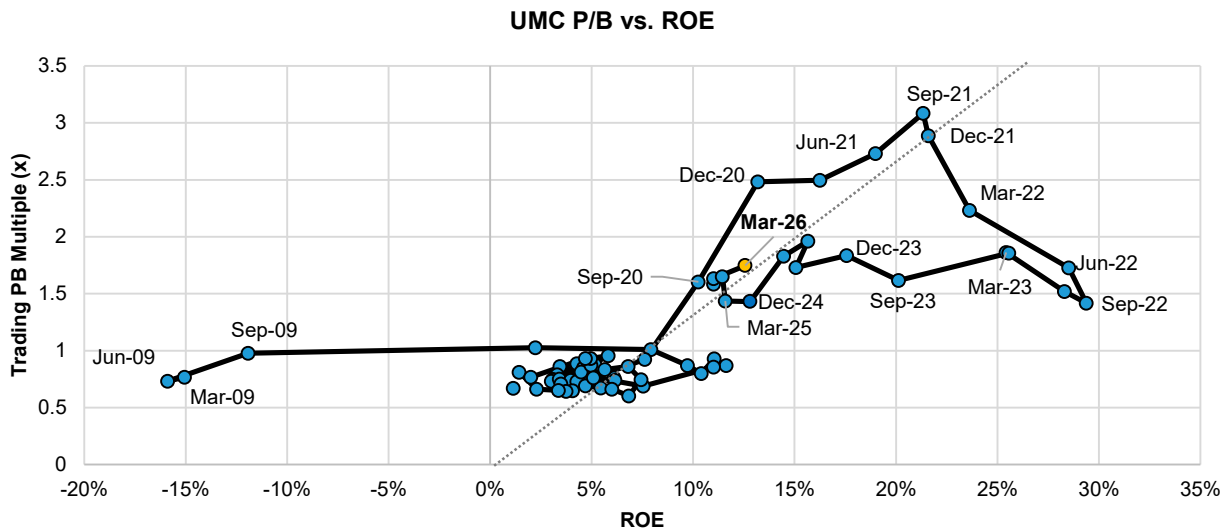
Source: Company report, Bernstein estimates and analysis

EXHIBIT 8: We model ROE to improve gradually & stay higher vs. pre-COVID level thanks to higher profit margin, but still shy of the COVID level.



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 9: Based on the historical correlation between P/B and ROE and our projected 2026-2028 ROE, we assign a 2.5x target P/B multiple for UMC.



Source: Bloomberg, Bernstein analysis

EXHIBIT 10: Our estimates are below consensus likely because we assume heavier depreciation across 2026-2028. We also assume the strong pricing of mature nodes will eventually drop due to competitive pressure in the long-term.

UMC	2026E					2027E					2028E		
	Bernstein New	Old	% Chg	Consensus	% Diff	Bernstein New	Old	% Chg	Consensus	% Diff	Bernstein New	Consensus	% Diff
Revenue (NT\$ B)	274.8	251.9	9.1%	275.3	-0.2%	315.9	267.4	18.1%	323.8	-2.4%	333.4	378.0	-11.8%
YoY	15.7%	6.0%		16.8%		14.9%	6.2%		17.6%		5.6%	16.7%	
Gross Margin	31.5%	27.8%	3.7%	31.5%	0.0%	33.2%	28.7%	4.4%	34.6%	-1.4%	36.7%	38.0%	-1.3%
Operating Margin	21.1%	17.1%	4.0%	21.1%	0.0%	23.1%	18.5%	4.6%	24.8%	-1.7%	26.5%	27.7%	-1.3%
EPS (NT\$)	4.58	3.31	38.4%	4.62	-0.8%	5.56	3.85	44.6%	5.74	-3.1%	6.62	7.82	-15.4%
YoY	37.2%	-0.9%		37.1%		21.4%	16.2%		24.3%		19.0%	36.2%	

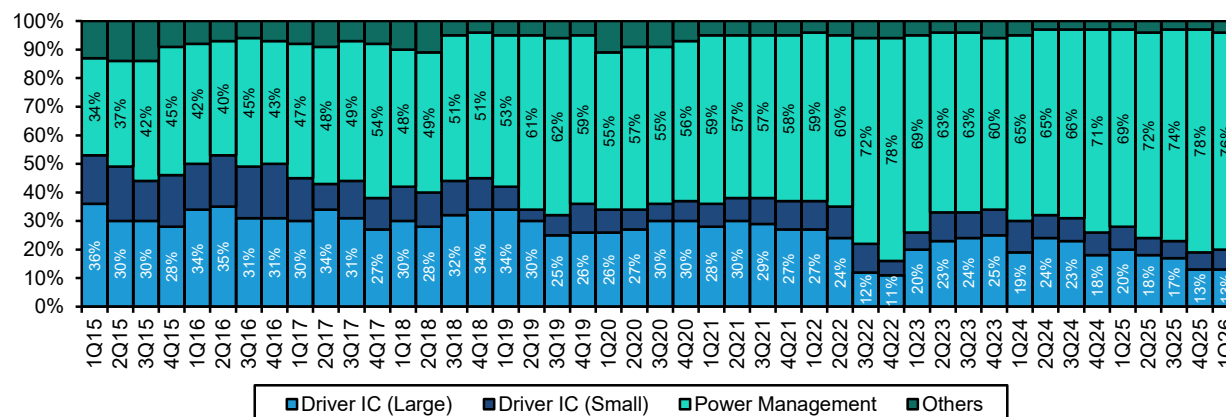
Source: Company reports, Bloomberg, Bernstein estimates and analysis

We maintain Market-Perform on Vanguard on its relatively better position, but see a new-term downside risk.

- Vanguard's stock price rose by c. 65% YTD on similar thesis as UMC's, but we believe Vanguard is relatively better positioned because of its large revenue exposure to power management product (Exhibit 11). The incremental interposer order from TSMC also improves the demand visibility for the Singapore 12-inch fab (VSMC). Its phase 1 capacity (now capped at 44KWPM, lowered from 55KWPM) is now fully secured by demand committed in LTAs. The fab successfully produced its first 40nm wafers in June & is on track for mass production in 1Q27. Given strong customer demand, the management is evaluating the feasibility of a phase 2.
- We revised up our utilization assumptions for VSMC and the strong AI power management IC demand for 0.18um node. We also lifted our ASP forecasts to reflect the above-average price of interposer, as well as the price adjustments on 8" mature nodes. Additionally, we modeled a lighter depreciation burden in outer years as TSMC bring its own equipment for interposer production. As a result, our EPS is revised up notably. We're now roughly in line with consensus on revenue across 2026-2028 but slightly below consensus on net earnings, mainly due to heavier opex assumptions (Exhibit 12 - Exhibit 15, Exhibit 18).
- We raise our target P/B to 4x from 3x on better visibility of VSMC and improved price outlook of 8" power management products. And with the revised Q5-Q8 BVPS, we increase our price target to **NT\$146.00**/share, implying 7% downside as of July 24 market close. Dividend yield will add c. 3% return. Maintain **Market-Perform** (Exhibit 16 - Exhibit 17).

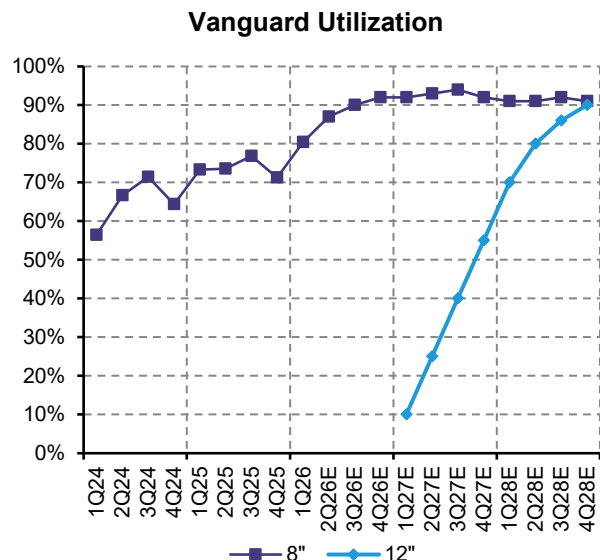
EXHIBIT 11: Vanguard has over 75% revenue exposure to Power Management products and thus relatively better-positioned, in our view.

Vanguard Revenue Breakdown by Platform



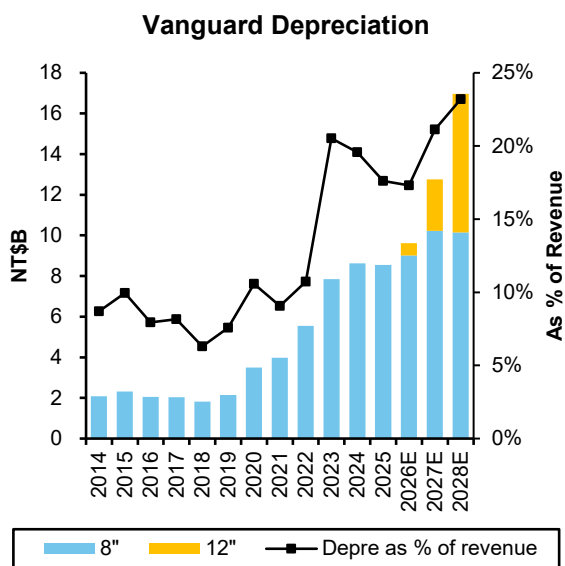
Source: Company reports and Bernstein analysis

EXHIBIT 12: We assumed a faster ramp up for the 12" new capacity at VSMC as interpose business from TSMC improves demand visibility.



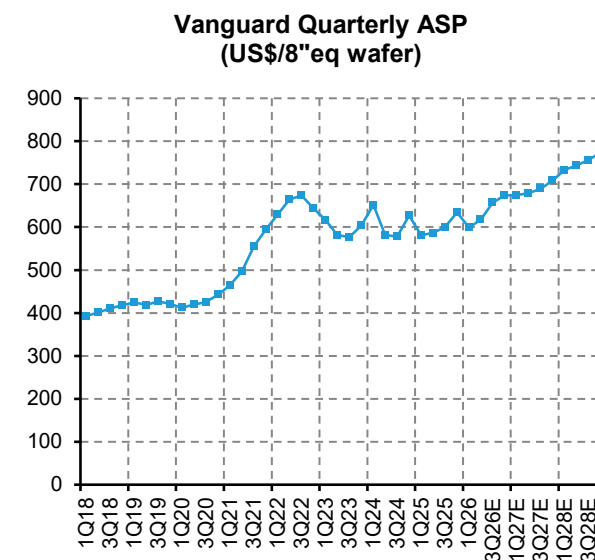
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 14: We model depreciation for VSMC to kick in from 4Q26 and rise further in 2027 & 2028.



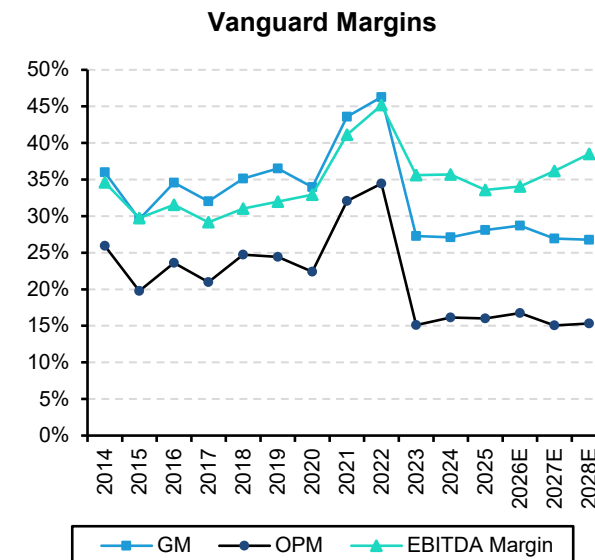
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 13: ASP is expected to go up as well given the favorable supply/demand dynamic in power management space and the interposer business.

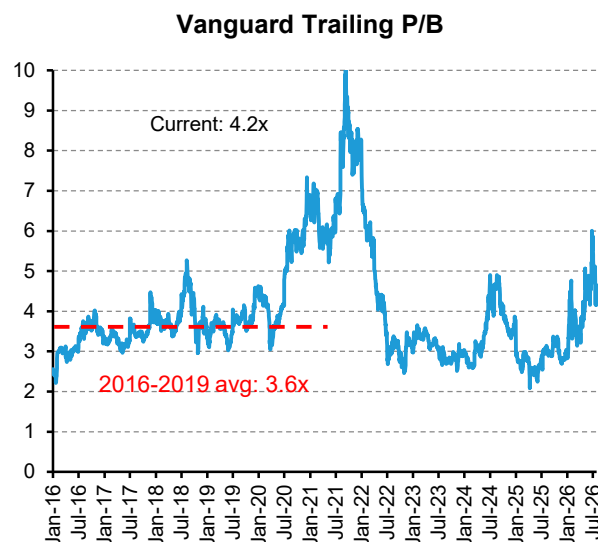


Source: Company reports, Bernstein estimates and analysis

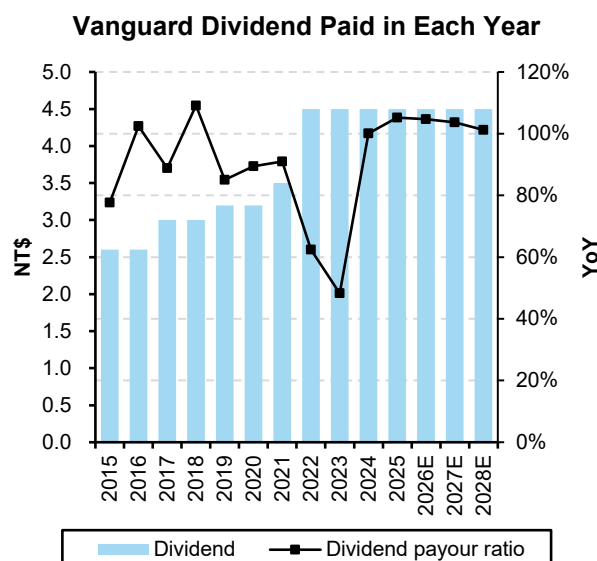
EXHIBIT 15: EBITDA margin is expected to improve from 2026 thanks to contribution from VSMC while GM & OPM are expected to be dragged by depreciation.



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 16: **Vanguard is now traded at 4.2x forward P/B.**

Source: Bloomberg, Bernstein analysis

EXHIBIT 17: **We expect Vanguard to maintain a stable dividend payout in terms of dollar amount across 2026-2028.**

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 18: **We're now roughly in line with consensus on revenue for 2026-2027, but slightly below consensus on net earnings, mainly due to heavier opex assumptions.**

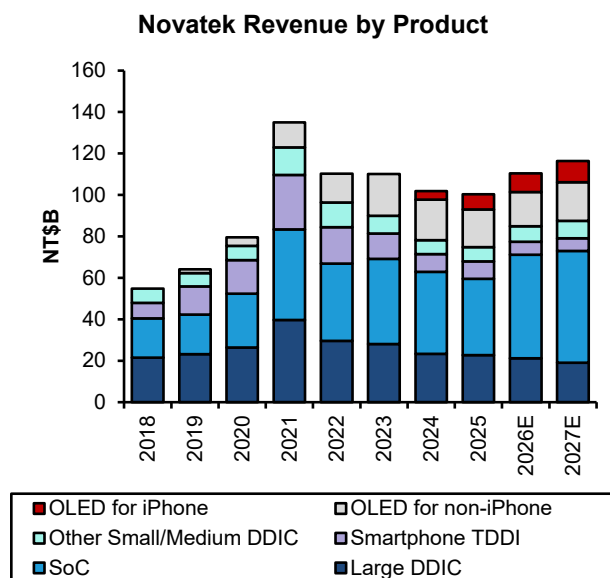
Vanguard	2026E					2027E					2028E		
	Bernstein New	Bernstein Old	% Chg	Consensus	% Diff	Bernstein New	Bernstein Old	% Chg	Consensus	% Diff	Bernstein New	Consensus	% Diff
Revenue (NT\$ M)	59,701	55,581	7.4%	58,232	2.5%	70,805	60,370	17.3%	73,398	-3.5%	88,847	94,325	-5.8%
YoY	22.9%	14.4%		20.3%		18.6%	8.6%		26.0%		25.5%	28.5%	
Gross Margin	32.1%	28.7%	3.4%	31.8%	0.3%	32.0%	26.9%	5.0%	30.4%	1.6%	32.3%	31.0%	1.3%
Operating Margin	19.4%	16.7%	2.7%	20.0%	-0.6%	18.4%	15.1%	3.3%	19.9%	-1.5%	20.1%	20.7%	-0.6%
EPS (NT\$)	5.70	4.34	31.3%	5.90	-3.5%	6.59	4.44	48.2%	7.26	-9.2%	8.95	9.19	-2.5%
YoY	32.5%	0.9%		37.0%		15.7%	2.4%		23.0%		35.9%	26.6%	

Source: Company reports, Bloomberg, Bernstein estimates and analysis

With no clear either upside or downside thesis, we maintain Market-Perform on Novatek.

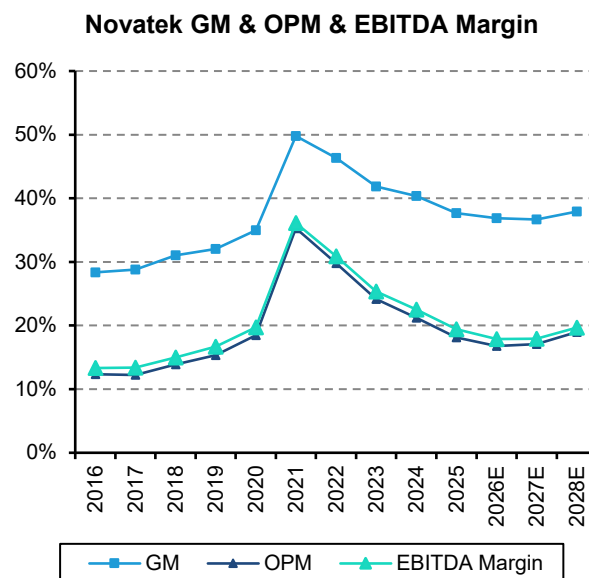
- We revise up the revenue growth of SoC segment as the edge AI video-related applications are emerging as a new growth driver for Novatek. Novatek also manages to increase its product price to reflect the cost increase from the memory embedded in Novatek's chips. Meanwhile, DDIC shipment may benefit from order pull-ins, but we worry about high memory cost to result in end demand destruction and consequently potential order cuts in 4Q26. We now model Novatek revenue to grow 9.7% in 2026 and c. 5% in 2027 & 2028. We raised our gross margin forecast as well to reflect better-than-expected cost pass-through and higher SoC contribution in the company revenue mix (Exhibit 19 - Exhibit 20).
- The stock is now trading at 16.8x forward P/E. We revised up our TP to **NT\$480.00** from NT\$370.00 based on higher Q5-Q8 earnings assumption and the same 14x target P/E multiple. It implies c. 7% downside in stock price. Dividend is likely to provide 4% yield. Maintain **Market-Perform** (Exhibit 21 - Exhibit 23).

EXHIBIT 19: We model SoC business to outperform in 2026 thanks to (1) new growth drivers in AI surveillance & machine vision, (2) higher ASP due to memory price.



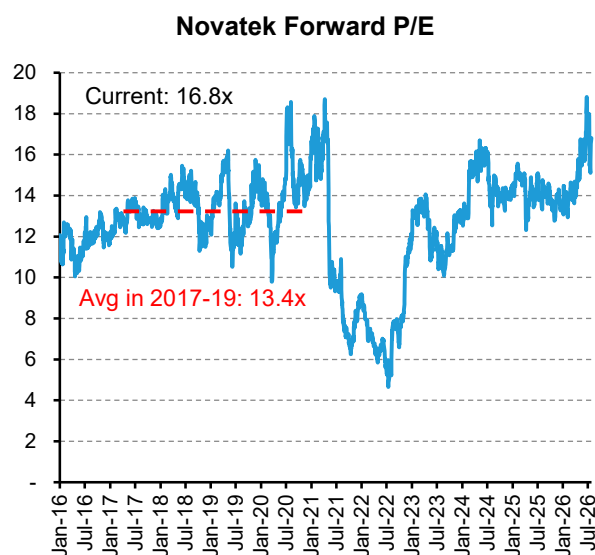
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 20: We model GM under pressure in 2026 & 1H27. Benefits from better product mix would be offset by raw material inflation.



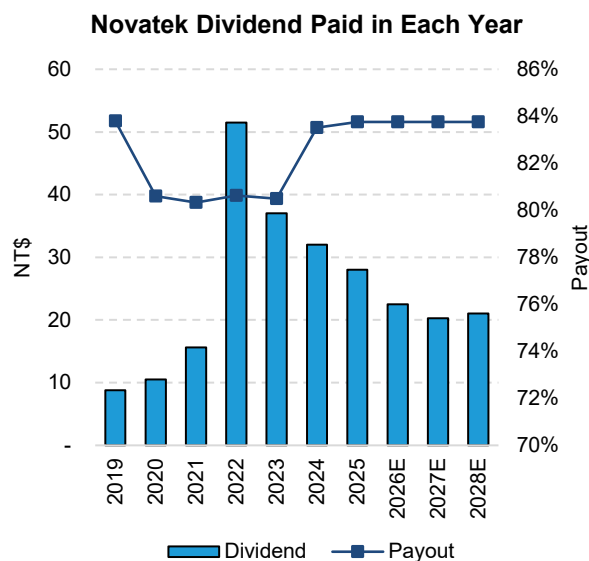
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 21: Novatek is now traded at 16.8x forward P/E.



Source: Bloomberg, Bernstein analysis

EXHIBIT 22: We assume dividend payout ratio stays at 85% for 2026-2028.



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 23: **We're roughly in line with consensus on topline and slightly above consensus on bottom line for 2026 & 2027.**

Novatek	2026E					2027E					2028E		
	Bernstein					Bernstein					Bernstein		
	New	Old	% Chg	Consensus	% Diff	New	Old	% Chg	Consensus	% Diff	New	Consensus	% Diff
Revenue (NT\$ M) YoY	110,418 9.7%	98,734 -1.9%	11.8%	100,390 -0.2%	10.0%	116,348 5.4%	100,648 1.9%	15.6%	107,986 7.6%	7.7%	122,389 5.2%	104,547 -3.2%	17.1%
Gross Margin	39.0%	36.9%	2.2%	37.8%	1.3%	39.0%	36.7%	2.3%	37.8%	1.2%	39.1%	37.9%	3.1%
Operating Margin	19.0%	16.8%	2.2%	18.0%	1.0%	19.3%	17.1%	2.3%	18.5%	0.8%	19.7%	18.9%	4.1%
EPS (NT\$) YoY	30.43 13.3%	24.21 -9.9%	25.7%	26.63 1.8%	14.3%	32.54 6.9%	25.10 3.7%	29.6%	28.76 8.0%	13.1%	34.89 7.2%	28.82 0.2%	21.1%

Source: Company reports, Bloomberg, Bernstein estimates and analysis

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 24: Financial forecast for UMC

NT\$ in Millions	2025				2026				2023	2024	2025	2026E	2027E	2028E
	Q1	Q2	Q3	Q4	Q1	Q2E	Q3E	Q4E						
Income Statement														
Net Sales	57,859	58,758	59,127	61,810	61,038	66,762	71,285	75,729	222,533	232,303	237,553	274,814	315,887	333,433
Gross Profit	15,447	16,878	17,623	18,958	17,818	20,433	22,975	25,406	77,744	75,654	68,906	86,632	104,779	122,483
SG&A Expense	2,159	2,274	2,343	2,448	2,523	2,612	2,880	3,013	10,572	9,749	9,223	11,029	12,686	13,680
R&D Expense	3,964	4,194	4,631	4,937	4,575	4,829	5,112	5,358	13,284	15,616	17,725	19,874	21,703	23,231
EBIT	9,786	10,820	11,118	12,225	11,276	13,461	15,512	17,631	57,891	51,613	43,949	57,880	72,980	88,238
EBITDA	23,914	25,326	26,112	27,855	27,263	29,866	32,410	35,129	98,375	99,780	103,208	124,669	140,648	151,646
Non-operating Income	(439)	(666)	3,526	3,278	5,367	1,425	1,125	933	13,022	4,607	5,699	8,852	10,771	11,423
Pretax Income	9,347	10,154	14,644	15,503	16,644	14,886	16,637	18,565	70,912	56,220	49,648	66,732	83,751	99,661
Tax	1,603	1,306	(300)	5,504	527	2,680	2,995	3,342	9,472	9,113	8,113	9,543	14,238	16,942
Net Income	7,743	8,848	14,944	9,999	16,117	12,207	13,642	15,223	61,440	47,106	41,535	57,189	69,513	82,718
Earnings Per Share (NT\$)	0.62	0.71	1.20	0.81	1.29	0.98	1.09	1.22	4.93	3.80	3.34	4.58	5.56	6.62
Margin Analysis														
Gross Profit	26.7%	28.7%	29.8%	30.7%	29.2%	30.6%	32.2%	33.5%	34.9%	32.6%	29.0%	31.5%	33.2%	36.7%
EBIT	16.9%	18.4%	18.8%	19.8%	18.5%	20.2%	21.8%	23.3%	26.0%	22.2%	18.5%	21.1%	23.1%	26.5%
EBITDA	41.3%	43.1%	44.2%	45.1%	44.7%	44.7%	45.5%	46.4%	44.2%	43.0%	43.4%	45.4%	44.5%	45.5%
Pretax Income	16.2%	17.3%	24.8%	25.1%	27.3%	22.3%	23.3%	24.5%	31.9%	24.2%	20.9%	24.3%	26.5%	29.9%
Net Income	13.4%	15.1%	25.3%	16.2%	26.4%	18.3%	19.1%	20.1%	27.6%	20.3%	17.5%	20.8%	22.0%	24.8%
Sequential Growth														
Net Sales	-4.2%	1.6%	0.6%	4.5%	-1.2%	9.4%	6.8%	6.2%	-20.2%	4.4%	2.3%	15.7%	14.9%	5.6%
Gross Profit	-15.8%	9.3%	4.4%	7.6%	-6.0%	14.7%	12.4%	10.6%	-38.2%	-2.7%	-8.9%	25.7%	20.9%	16.9%
EBIT	-18.2%	11%	2.8%	10.0%	-7.8%	19%	15.2%	13.7%	-44.5%	-10.8%	-14.8%	31.7%	26.1%	20.9%
EBITDA	-5.9%	5.9%	3.1%	6.7%	-2.1%	9.5%	8.5%	8.4%	-33.7%	1.4%	3.4%	20.8%	12.8%	7.8%
Pretax	-11.1%	9%	44.2%	5.9%	7.4%	-11%	11.8%	12%	-33.2%	-20.7%	-11.7%	34.4%	25.5%	19.0%
Net Income	-8.5%	14%	68.9%	-33.1%	61.2%	-24%	11.8%	12%	-30.2%	-23.3%	-11.8%	37.7%	21.6%	19.0%
Earnings Per Share	-8.8%	14.5%	69.0%	-32.5%	59.3%	-24.2%	11.8%	11.6%	-30.4%	-23.0%	-12.1%	37.2%	21.4%	19.0%
Balance Sheet														
Cash and Cash Equivalents									132,554	105,000	110,660	150,106	199,081	260,590
Accounts Receivable									29,586	33,343	31,274	40,300	43,257	44,050
Inventory									35,713	35,782	37,228	44,973	47,032	46,350
Net Fixed Assets									239,123	279,059	271,395	251,746	225,028	196,270
Total Assets									559,187	570,201	578,996	628,268	660,482	700,300
Accounts Payable									7,526	7,633	9,170	10,033	10,706	10,765
Total Debt									80,166	81,462	78,970	74,221	73,252	71,894
Total Equity									359,579	378,185	379,855	416,080	448,120	485,870
Cash Flow														
Cash Flow From Operations									86,000	93,872	99,864	132,107	131,859	146,192
Cash Flow From Investments									(97,787)	(85,941)	(53,154)	(56,843)	(44,442)	(38,357)
Cash Flow From Financing									(29,086)	(39,200)	(39,204)	(37,229)	(38,442)	(46,327)
Depreciation & Amortization									40,484	48,168	59,259	66,789	67,668	63,409
Capex									(91,474)	(88,544)	(47,745)	(47,176)	(40,950)	(34,650)
Key Financial Metrics														
Return on Common Equity									17.6%	12.8%	11.0%	14.4%	16.1%	17.7%
Return on Assets									11.2%	8.4%	7.3%	9.5%	10.8%	12.2%
Capex Intensity									41%	38%	20%	17%	13%	10%
Asset Turnover									0.4x	0.4x	0.4x	0.4x	0.5x	0.5x
Debt / Equity									22%	22%	21%	18%	16%	15%
Net Debt / Equity									-15%	-6%	-8%	-18%	-28%	-39%
Debt / Capital									18%	18%	17%	15%	14%	13%
Dividend Per Share (NT\$)									3.6	3.0	2.9	2.5	3.0	3.6

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 25: Financial forecast for Vanguard

NT\$ in Millions	2025				2026				2024	2025	2026E	2027E	2028E
	Q1	Q2	Q3	Q4	Q1	Q2E	Q3E	Q4E					
Income Statement													
Net Sales	11,949	11,699	12,349	12,594	12,532	14,396	15,975	16,798	44,055	48,591	59,701	70,805	88,847
Gross Profit	3,597	3,280	3,312	3,465	3,671	4,705	5,260	5,544	11,933	13,654	19,179	22,629	28,713
SG&A Expense	838	861	915	1,075	974	1,065	1,198	1,260	2,740	3,691	4,498	5,210	5,732
R&D Expense	521	522	565	582	610	633	895	941	2,081	2,190	3,079	4,418	5,118
EBIT	2,238	1,897	1,832	1,807	2,087	3,006	3,167	3,343	7,111	7,773	11,603	13,001	17,863
EBITDA	4,284	3,983	4,037	4,022	4,278	5,256	5,759	5,927	15,733	16,326	21,221	25,945	34,272
Non-operating Income	437	485	267	288	350	174	96	357	1,641	1,477	978	1,678	2,173
Pretax Income	2,675	2,382	2,099	2,095	2,437	3,181	3,263	3,701	8,752	9,251	12,580	14,679	20,036
Tax	269	369	414	429	273	636	587	666	1,705	1,480	2,162	2,553	3,486
Net Income	2,406	2,013	1,685	1,666	2,164	2,544	2,675	3,035	7,047	7,770	10,418	12,126	16,551
Earnings Per Share (NT\$)	1.31	1.11	0.93	0.95	1.22	1.38	1.45	1.64	4.28	4.30	5.70	6.59	8.95
Margin Analysis													
Gross Profit	30.1%	28.0%	26.8%	27.5%	29.3%	32.7%	32.9%	33.0%	27.1%	28.1%	32.1%	32.0%	32.3%
EBIT	18.7%	16.2%	14.8%	14.3%	16.7%	20.9%	19.8%	19.9%	16.1%	16.0%	19.4%	18.4%	20.1%
EBITDA	35.9%	34.0%	32.7%	31.9%	34.1%	36.5%	36.1%	35.3%	35.7%	33.6%	35.5%	36.6%	38.6%
Pretax Income	22.4%	20.4%	17.0%	16.6%	19.4%	22.1%	20.4%	22.0%	19.9%	19.0%	21.1%	20.7%	22.6%
Net Income	20.1%	17.2%	13.6%	13.2%	17.3%	17.7%	16.7%	18.1%	16.0%	16.0%	17.5%	17.1%	18.6%
Sequential Growth													
Net Sales	3.4%	-2.1%	5.6%	2.0%	-0.5%	14.9%	11.0%	5.2%	15.1%	10.3%	22.9%	18.6%	25.5%
Gross Profit	8.5%	-8.8%	1.0%	4.6%	6.0%	28.2%	11.8%	5.4%	14.2%	14.4%	40.5%	18.0%	26.9%
EBIT	16.3%	-15.3%	-3.4%	-1.4%	15.5%	44.1%	5.3%	5.6%	23.2%	9.3%	49.3%	12.0%	37.4%
EBITDA	2.5%	-7.0%	1.4%	-0.4%	6.4%	22.9%	9.6%	2.9%	15.5%	3.8%	30.0%	22.3%	32.1%
Pretax	14.6%	-11.0%	-11.9%	-0.2%	16.3%	30.5%	2.6%	13.4%	-0.9%	5.7%	36.0%	16.7%	36.5%
Net Income	30.5%	-16.4%	-16.3%	-1.1%	29.8%	17.6%	5.1%	13.4%	-4.4%	10.3%	34.1%	16.4%	36.5%
Earnings Per Share	19.0%	-15.4%	-16.6%	2.6%	28.5%	12.8%	5.2%	13.4%	-4.9%	0.6%	32.5%	15.7%	35.9%
Balance Sheet													
Cash and Cash Equivalents									60,695	39,635	85,494	117,566	164,878
Accounts Receivable									6,235	6,748	9,766	9,978	13,415
Inventory									6,221	6,937	7,340	7,775	9,816
Net Fixed Assets									44,141	113,869	107,377	99,843	66,834
Total Assets									148,706	200,469	245,997	273,486	298,098
Accounts Payable									1,416	1,724	2,318	2,678	3,381
Total Debt									24,656	31,939	37,636	36,432	33,329
Total Equity									68,667	83,590	107,180	129,174	143,267
Cash Flow													
Cash Flow From Operations									24,731	25,439	31,794	28,691	36,148
Cash Flow From Investments									(11,699)	(59,711)	(11,501)	(5,283)	16,725
Cash Flow From Financing									17,530	13,460	25,437	8,664	(5,561)
Depreciation & Amortization									8,622	8,552	9,618	12,944	16,409
Capex									(15,909)	(63,265)	(66,292)	(25,410)	(3,400)
Key Financial Metrics													
Return on Common Equity									12.8%	12.4%	16.0%	17.5%	21.9%
Return on Assets									5.6%	4.5%	4.7%	4.7%	5.8%
Capex Intensity									36.1%	130.2%	111.0%	35.9%	3.8%
Asset Turnover									0.3x	0.2x	0.2x	0.3x	0.3x
Debt / Equity									35.9%	38.2%	35.1%	28.2%	23.3%
Net Debt / Equity									-52%	-9%	-45%	-63%	-92%
Debt / Capital									26.4%	27.6%	26.0%	22.0%	18.9%
Dividend Per Share (NT\$)									4.5	4.5	4.5	4.5	4.5

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 26: **Financial forecast for Novatek**

NT\$ in Millions	2025				2026				2023	2024	2025	2026E	2027E	2028E
	Q1	Q2	Q3	Q4	Q1	Q2E	Q3E	Q4E						
Income Statement														
Net Sales	27,120	26,152	24,574	22,817	23,145	28,381	29,691	29,201	110,429	102,788	100,663	110,418	116,348	122,389
Gross Profit	10,783	9,494	8,919	8,713	9,040	11,148	11,583	11,342	46,212	41,472	37,908	43,113	45,318	47,835
SG&A Expense	784	756	716	691	736	823	831	832	2,759	2,859	2,946	3,223	3,378	3,541
R&D Expense	4,303	3,915	4,346	4,160	4,276	4,939	4,880	4,838	16,791	16,807	16,725	18,933	19,440	20,156
EBIT	5,696	4,823	3,857	3,861	4,028	5,387	5,871	5,672	26,662	21,806	18,237	20,957	22,500	24,138
EBITDA	6,017	5,144	4,192	4,219	4,389	5,697	6,184	5,977	28,001	23,120	19,573	22,247	23,623	25,186
Non-operating Income	445	(221)	518	554	362	254	332	252	1,569	2,320	1,296	1,200	1,214	1,287
Pretax Income	6,141	4,602	4,375	4,415	4,389	5,640	6,204	5,924	28,231	24,126	19,534	22,157	23,714	25,426
Tax	877	863	720	726	622	1,015	1,024	977	4,914	3,784	3,186	3,639	3,913	4,195
Net Income	5,264	3,739	3,655	3,690	3,767	4,625	5,180	4,946	23,318	20,342	16,348	18,519	19,801	21,230
Earnings Per Share (NT\$)	8.65	6.14	6.01	6.06	6.19	7.60	8.51	8.13	38.32	33.43	26.87	30.43	32.54	34.89
Margin Analysis														
Gross Profit	39.8%	36.3%	36.3%	38.2%	39.1%	39.3%	39.0%	38.8%	41.8%	40.3%	37.7%	39.0%	39.0%	39.1%
EBIT	21.0%	18.4%	15.7%	16.9%	17.4%	19.0%	19.8%	19.4%	24.1%	21.2%	18.1%	19.0%	19.3%	19.7%
EBITDA	22.2%	19.7%	17.1%	18.5%	19.0%	20.1%	20.8%	20.5%	25.4%	22.5%	19.4%	20.1%	20.3%	20.6%
Pretax Income	22.6%	17.6%	17.8%	19.4%	19.0%	19.9%	20.9%	20.3%	25.6%	23.5%	19.4%	20.1%	20.4%	20.8%
Net Income	19.4%	14.3%	14.9%	16.2%	16.3%	16.3%	17.4%	16.9%	21.1%	19.8%	16.2%	16.8%	17.0%	17.3%
SG&A as % of Sales	2.9%	2.9%	2.9%	3.0%	3.2%	2.9%	2.8%	2.9%	2.5%	2.8%	2.9%	2.9%	2.9%	2.9%
R&D as % of Sales	15.9%	15.0%	17.7%	18.2%	18.5%	17.4%	16.4%	16.6%	15.2%	16.4%	16.6%	17.1%	16.7%	16.5%
Sequential Growth														
Net Sales	7.4%	-3.6%	-6.0%	-7.1%	1.4%	22.6%	4.6%	-1.6%	0.4%	-6.9%	-2.1%	9.7%	5.4%	5.2%
Gross Profit	8.0%	-12.0%	-6.1%	-2.3%	3.8%	23.3%	3.9%	-2.1%	-9.3%	-10.3%	-8.6%	13.7%	5.1%	5.6%
EBIT	15.9%	-15.3%	-20.0%	0.1%	4.3%	33.7%	9.0%	-3.4%	-18.5%	-18.2%	-16.4%	14.9%	7.4%	7.3%
EBITDA	14.7%	-14.5%	-18.5%	0.6%	4.0%	29.8%	8.5%	-3.4%	-17.5%	-17.4%	-15.3%	13.7%	6.2%	6.6%
Pretax	7.9%	-25.1%	-4.9%	0.9%	-0.6%	28.5%	10.0%	-4.5%	-18.2%	-14.5%	-19.0%	13.4%	7.0%	7.2%
Net Income	9.6%	-29.0%	-2.3%	1.0%	2.1%	22.8%	12.0%	-4.5%	-16.6%	-12.8%	-19.6%	13.3%	6.9%	7.2%
Earnings Per Share	9.6%	-29.0%	-2.3%	1.0%	2.1%	22.8%	12.0%	-4.5%	-16.6%	-12.8%	-19.6%	13.3%	6.9%	7.2%
Balance Sheet														
Cash and Cash Equivalents									53,889	48,771	42,419	54,728	56,172	59,964
Accounts Receivable									16,935	20,274	21,099	22,883	27,055	27,566
Inventory									8,551	8,701	8,641	10,240	11,382	11,681
Net Fixed Assets									6,876	7,023	7,210	3,088	2,779	2,588
Total Assets									104,443	102,401	97,366	109,660	116,248	120,727
Accounts Payable									12,579	10,635	9,352	12,803	13,460	13,665
Total Debt									373	455	552	67	78	82
Total Equity									67,217	67,923	67,739	72,987	77,276	81,922
Cash Flow														
Cash Flow From Operations									29,981	16,652	13,629	23,537	17,758	21,229
Cash Flow From Investments									(1,702)	(849)	(1,232)	2,969	(814)	(857)
Cash Flow From Financing									(23,552)	(20,956)	(18,741)	(14,216)	(15,499)	(16,581)
Depreciation & Amortization									1,340	1,314	1,336	1,290	1,123	1,048
Capex									(897)	(1,276)	(1,211)	2,931	(814)	(857)
Key Financial Metrics														
Return on Common Equity									35.1%	30.1%	24.1%	26.3%	26.4%	26.7%
Return on Assets									22.8%	19.7%	16.4%	17.9%	17.5%	17.9%
Debt / Equity									0.6%	0.7%	0.8%	0.1%	0.1%	0.1%
Net Debt / Equity									-80%	-71%	-62%	-75%	-73%	-73%
Debt / Capital									0.6%	0.7%	0.8%	0.1%	0.1%	0.1%
Dividend Per Share (NT\$)									37.0	32.0	28.0	22.5	25.5	27.3

Source: Company reports, Bernstein estimates and analysis

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Taiwan Semiconductor Manufacturing Co Ltd**

We set a NT\$2,780 1-year target price using a P/E multiple of 20x against our forward Q5-Q8 EPS estimate of NT\$139. We derive our PT for secondary ticker (TSM.US) by dividing the primary PT by FX and arrive at US\$430.

United Microelectronics Corp

We set a NT\$88 1-year target price using a P/B multiple of 2.5x against our forward Q5-Q8 BVPS estimate of NT\$35.2. We derive our PT for secondary ticker (UMC.US) by dividing the primary PT by FX and arrive at US\$13.6.

Vanguard International Semiconductor Corp

We set a NT\$146 1-year target price using a P/B multiple of 4x against our forward Q5-Q8 BVPS estimate of NT\$36.5.

Novatek Microelectronics Corp

We set a NT\$480 1-year price target using a P/E multiple of 14x against our forward Q5-Q8 EPS estimate of NT\$34.

RISKS**Taiwan Semiconductor Manufacturing Co Ltd**

Downside risks are: (1) Market-wide multiple contraction. (2) Intel regains and keeps technology superiority. (3) Geopolitical uncertainties

United Microelectronics Corp

There are a number of upside risks to our UMC target price: (1) Better-than-expected progress/pricing in 28nm; (2) Better-than-expected end demand recovery; (3) Lighter-than-expected competitive pressure from Chinese rivals.

Vanguard International Semiconductor Corp

There are a number of risks to our Vanguard target price: (1) Faster or slower integration and cost reduction of the Singapore fab; (2) Faster- or slower-than-expected demand of driver IC business and migration to 12"; (3) Exchange rate fluctuation poses substantial risk to its revenue and profit; (4) Changes in China's government semiconductor support or policy.

Novatek Microelectronics Corp

There are a number of upside/downside risks to our Novatek price target: (1) Slower or faster shipment growth in OLED smartphones (2) ASP & margin erosion or share loss in TDDI (3) Unexpected entry of competitors, particularly from China (4) Slower or faster than expected progress in fingerprint sensor product

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Neutral (C-N): The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
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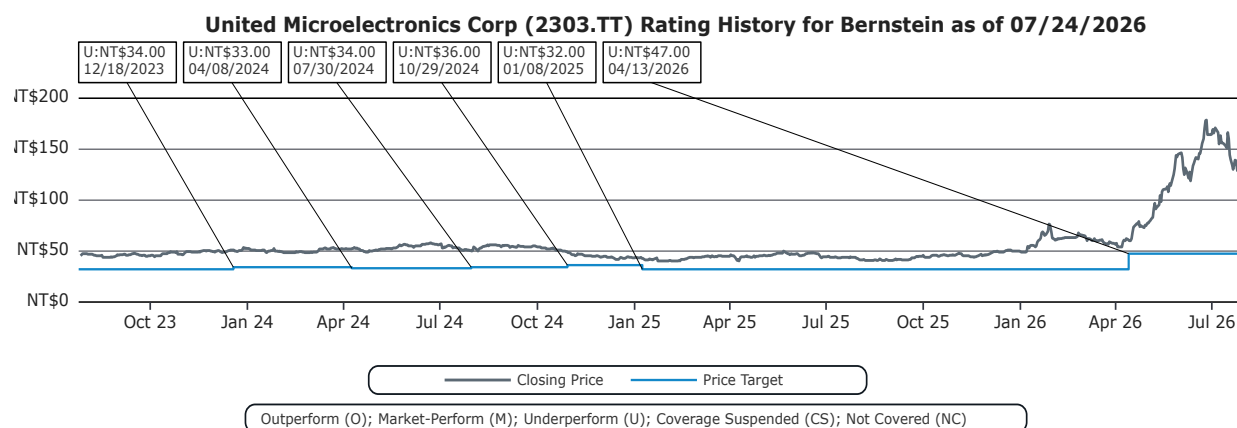
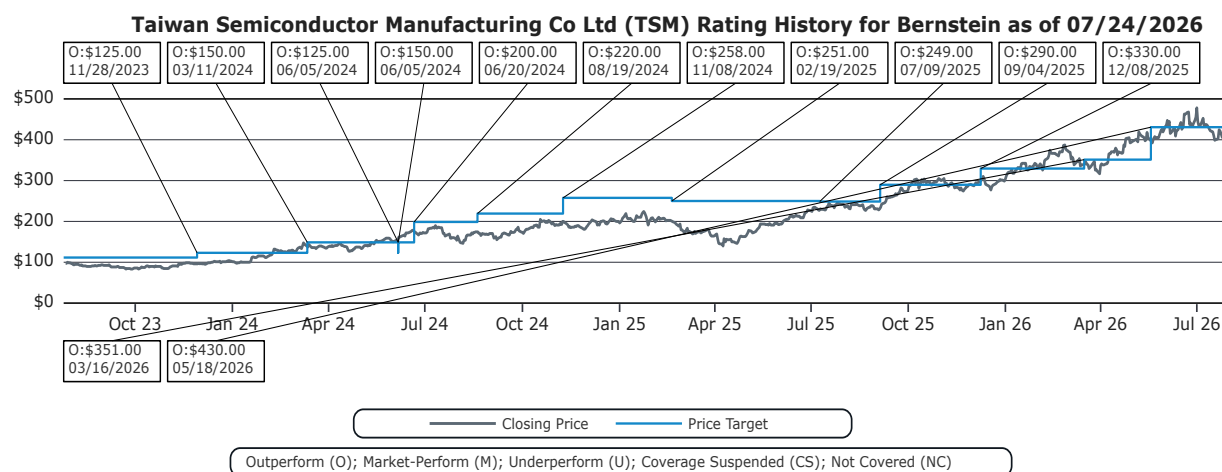
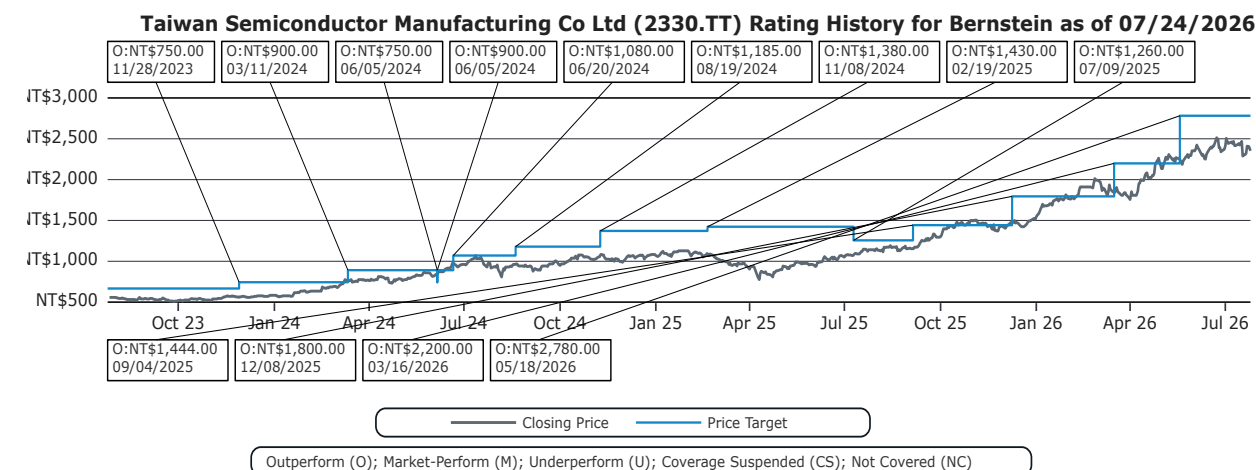
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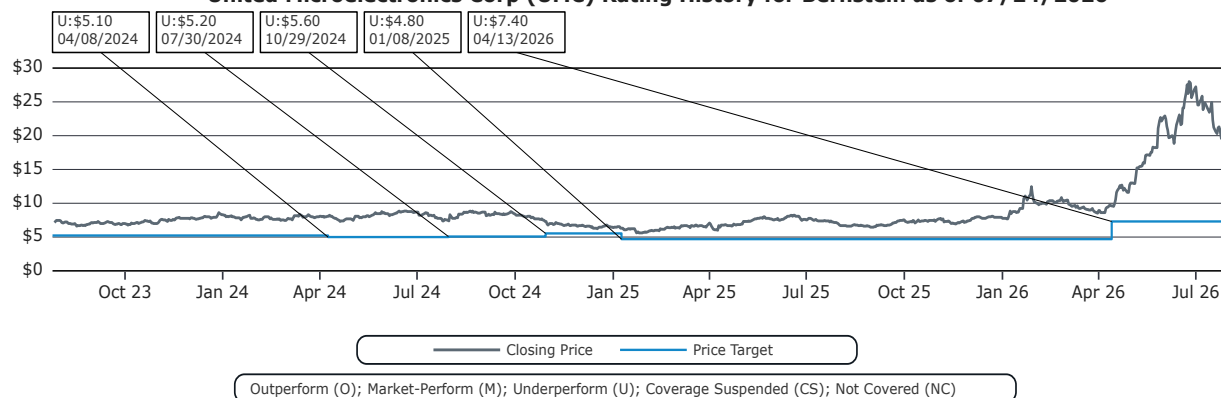
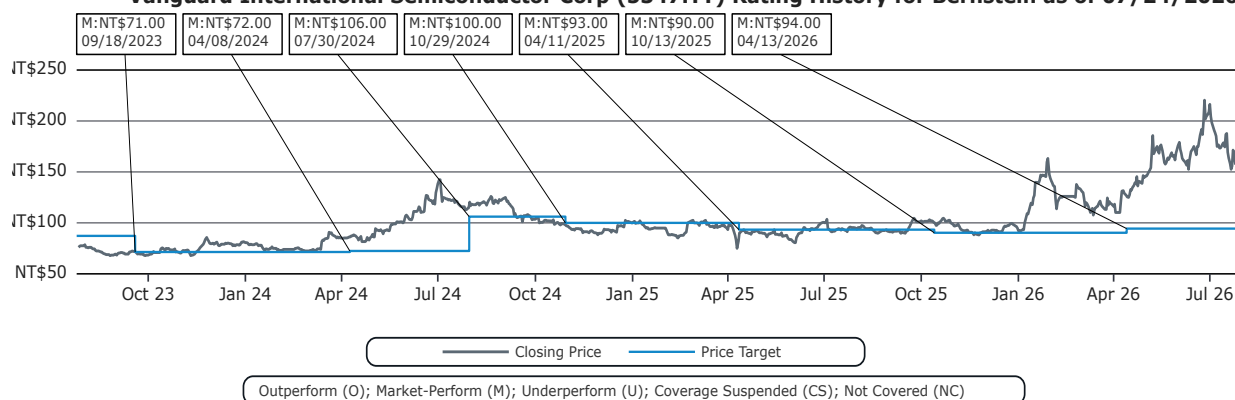
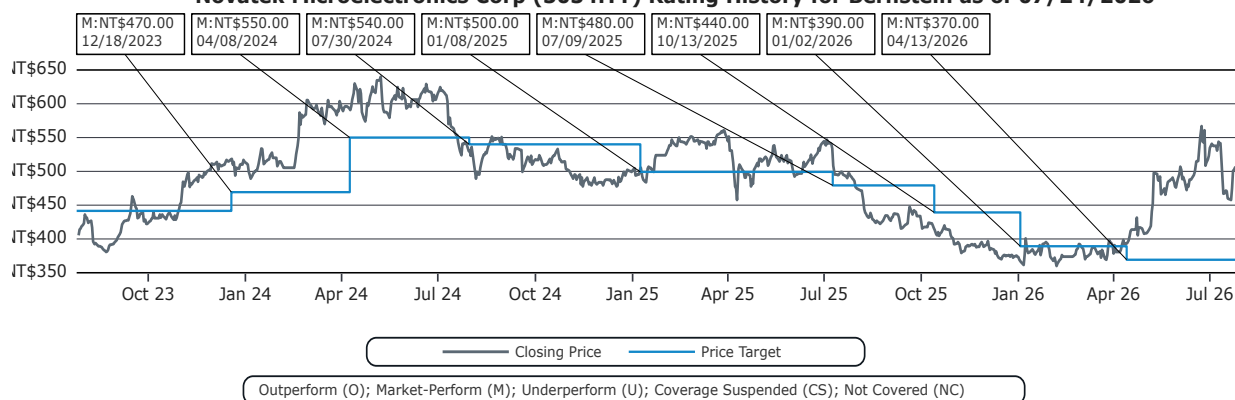
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Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY



United Microelectronics Corp (UMC) Rating History for Bernstein as of 07/24/2026**Vanguard International Semiconductor Corp (5347.TT) Rating History for Bernstein as of 07/24/2026****Novatek Microelectronics Corp (3034.TT) Rating History for Bernstein as of 07/24/2026**

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